Project1:

Data Analysis Plan

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1. Introduction

This project is a secondary analysis of data from the Multicenter AIDS Cohort Study (MACS), an ongoing prospective cohort study of HIV-1 infection in homosexual and bisexual men in four major U.S. cities. The dataset contains up to 8 years of annual follow-up on 715 HIV-infected men after initiating HAART (highly active antiretroviral therapy), including longitudinal laboratory measures, quality-of-life outcomes, and demographic and health information.

The primary objective of this analysis is to determine whether treatment response two years after initiating HAART differs between men who reported using hard drugs at baseline and those who did not report such use. This question was biologically motivated by previous studies in vitro and in animal models that suggest hard drug use may suppress immune function and increase HIV replication. However, evidence from human studies has been inconsistent.

Treatment response in this study will be evaluated using four outcome measures assessed two years after HAART initiation. The first outcome is viral load defined as the number of HIV RNA copies per milliliter of blood. The second outcome is CD4+ T-cell count, a measure of immune function. Together, these laboratory measures provide objective indicators of immunity suppression.

In addition to laboratory outcomes, two quality-of-life (QOL) measures derived from the SF-36 instrument will be examined. The aggregate physical health score and the aggregate mental health score, each ranging from 0 to 100, with higher scores indicating better perceived health.

The purpose of this preliminary report is to develop an analysis plan to answer the specific hypothesis guiding this analysis, that men who reported hard drug use at baseline will demonstrate a poorer treatment response at two years compared to non-users. Poorer response would be reflected by higher viral load, lower CD4+ T-cell count, and potentially lower physical and mental QOL scores.

2. Methods

2.1 Data Cleaning

The primary exposure variable, baseline hard drug use, was coded as a binary indicator (Yes versus No). Comorbidity variables including diabetes, hypertension (HBP), kidney disease, and liver disease (stage 3/4) contained additional categories reflecting “trajectory-based” classifications and “insufficient data.” For the purposes of exploratory analyses,

trajectory-based “Yes” and “No” categories were collapsed into single “Yes” and “No” levels, and observations coded as “insufficient data” were treated as missing or NA values. However, because these variables were not included in the primary models, this recoding did not affect the primary results.

Self-reported race/ethnicity will be included as a covariate to account for potential confounding by social drivers of health and healthcare access disparities. For analyses, we will combine White-Hispanic, Black-Hispanic, and Other-Hispanic as one category for Hispanic. We acknowledge that race is a social construct and that observed differences likely reflect structural inequities rather than biological differences. Additionally, we are not estimating race effects but purely adjusting for confounding.

Viral load values were highly right-skewed and were transformed using logarithm with base-10 for modeling. CD4+ T-cell counts were examined for distributional properties using histograms and residual diagnostics from preliminary regression models. Although the raw CD4 distribution was moderately right-skewed, residual diagnostics indicated acceptable normality and consistent variance. Therefore, CD4 counts will be retained on the original scale.

2.2 Data Analysis

The primary objective of this analysis is to evaluate whether baseline hard drug use is associated with treatment response two years after initiation of HAART. Treatment response will be assessed separately for viral load, CD4+ T-cell count, physical quality-of-life score, and mental quality-of-life score. A secondary objective is to assess whether treatment adherence mediates this relationship. Adherence was measured during follow-up after HAART initiation and is hypothesized to lie on the causal pathway between baseline hard drug use and treatment response.

For each outcome, analyses will be conducted using general linear regression to assess the association between baseline hard drug use and treatment response, while adjusting for relevant covariates and potential confounders. Both models adjusted for age, BMI, smoking status, education, race/ethnicity, and baseline disease severity.

For CD4+ T-cell count, physical quality-of-life score, and mental quality-of-life score, outcomes will be modeled on their original scales. Each model will include the corresponding baseline value of the outcome as well as the same set of prespecified covariates described above.

In addition to frequentist linear regression models, Bayesian regression models will also be fit for each outcome to allow comparison of inference under a Bayesian framework. The same model structure will be used in both frameworks. Under the Bayesian prior

specification, weakly informative normal priors will be placed on regression coefficients, and a half-Cauchy prior will be placed on the residual standard deviation.

Mediation will be evaluated by using a two-model approach. First, adherence will be modeled as a function of baseline hard drug use and baseline covariates. Second, the Year 2 outcome will be modeled as a function of baseline hard drug use, adherence, baseline outcome level, and covariates. Bayesian mediation models were specified using joint regression of adherence and Year 2 outcomes. Indirect effects were computed as the posterior distribution of the product of the exposure–mediator and mediator–outcome regression coefficients. Highest density intervals were used to assess evidence of mediation.

Model assumptions for the frequentist analyses will be evaluated using residual diagnostic plots, including assessment of linearity, constant variance, and normality of residuals. Statistical significance under the frequentist framework will be determined using two-sided hypothesis tests with an alpha level of 0.05. P-values less than 0.05 will be considered statistically significant.

Under the Bayesian framework, posterior distributions obtained from Markov Chain Monte Carlo (MCMC) sampling will be used to assess significance. Posterior means and 95% highest density intervals (HDIs) will be reported for each parameter estimate. Convergence of each Bayesian model will be assessed using trace plots and the Gelman–Rubin statistic ($\hat{R}$).

To assess the potential impact of loss to follow-up, baseline characteristics will be compared between participants with complete baseline and Year 2 measurements and those without Year 2 data. Differences in continuous variables will be evaluated using Welch's t-test or Wilcoxon rank sum test, and differences in categorical variables will be assessed using Pearson's chi-square tests or Fisher's exact tests. Findings will be interpreted under the assumption that missingness is at least missing at random (MAR), where dropout depends on observed variables.

All regression analyses will be conducted using complete-case observations for the variables included in each specific model. The number of observations removed due to missingness will be reported alongside model results.

3. Preliminary Results

Characteristic	N	Overall[1,2] N = 715	Hard Druge Use: No[1,2] N = 649	Hard Druge Use: Yes[1,2] N = 66
Mental QOL	713			
Mean (SD)		45.4 (13.3)	45.6 (13.4)	43.4 (12.4)
Median (Q1, Q3)		49.3 (36.8, 56.2)	49.7 (36.8, 56.3)	46.6 (31.9, 54.2)
Min - Max		7.2 - 69.8	7.2 - 69.8	20.8 - 59.6
NA		2 (0.3%)	2 (0.3%)	0 (0%)
Physical QOL	713			
Mean (SD)		50.1 (9.5)	50.7 (9.4)	44.8 (9.6)
Median (Q1, Q3)		52.8 (44.4, 56.7)	53.3 (45.2, 56.9)	45.3 (38.8, 52.7)
Min - Max		19.2 - 69.0	19.2 - 69.0	27.0 - 62.9
NA		2 (0.3%)	2 (0.3%)	0 (0%)
Viral Load (log_{10} copies/mL)	673			
Mean (SD)		4.5 (0.9)	4.5 (0.9)	4.7 (0.8)
Median (Q1, Q3)		4.5 (4.0, 5.1)	4.5 (4.0, 5.1)	4.7 (4.0, 5.5)
Min - Max		0.2 - 8.3	0.2 - 8.3	2.9 - 6.4
NA		42 (5.9%)	39 (6.0%)	3 (4.5%)
CD4 T-cell Count	691			
Mean (SD)		380.3 (209.5)	385.0 (211.3)	335.1 (187.0)
Median (Q1, Q3)		360.5 (229.7, 506.6)	362.7 (236.1, 506.6)	300.6 (188.8, 464.9)
Min - Max		10.9 - 1,217.5	10.9 - 1,217.5	10.9 - 778.0
NA		24 (3.4%)	24 (3.7%)	0 (0%)
Age	715			
Mean (SD)		42.6 (9.4)	42.4 (9.4)	43.7 (9.9)
Median (Q1, Q3)		42.0 (36.0, 49.0)	42.0 (36.0, 49.0)	45.5 (35.0, 52.0)
Min - Max		19.0 - 73.0	19.0 - 73.0	26.0 - 61.0
BMI	688			
Mean (SD)		34.8 (94.4)	36.1 (99.2)	22.4 (6.8)
Median (Q1, Q3)		24.5 (22.1, 27.7)	24.7 (22.4, 27.7)	22.3 (19.8, 25.9)
Min - Max		-1.0 - 999.0	-1.0 - 999.0	-1.0 - 36.8
NA		27 (3.8%)	27 (4.2%)	0 (0%)
Smoke	715			
Never		188 (26%)	187 (29%)	1 (1.5%)
Former		223 (31%)	211 (33%)	12 (18%)
Current		304 (43%)	251 (39%)	53 (80%)
Education	715			
College		394 (55%)	371 (57%)	23 (35%)
High School or less		180 (25%)	147 (23%)	33 (50%)
Post-grad		141 (20%)	131 (20%)	10 (15%)
Race	715			
Black non-Hisp		208 (29%)	180 (28%)	28 (42%)
Hispanic		85 (12%)	79 (12%)	6 (9.1%)
White non-Hisp		422 (59%)	390 (60%)	32 (48%)
Depression Scale (>=16 is depressed)	694			
Mean (SD)		14.7 (12.3)	14.2 (12.2)	19.3 (12.4)
Median (Q1, Q3)		11.0 (5.0, 23.0)	11.0 (4.0, 22.0)	18.0 (10.0, 27.0)
Min - Max		0.0 - 51.0	0.0 - 51.0	1.0 - 47.0
NA		21 (2.9%)	21 (3.2%)	0 (0%)

[1] n (%)
[2] Table includes all subjects from cohort at baseline (years=0)

Table 1. *Baseline characteristics of HIV-infected men initiating HAART, overall and stratified by baseline hard drug use.*

Baseline characteristics of the study cohort are presented in Table 1. At HAART initiation, 9.2% of participants reported hard drug use. Drug users were typically more depressed, had lower baseline CD4 counts, and had comparable baseline viral load levels and QOL metrics relative to non-users.

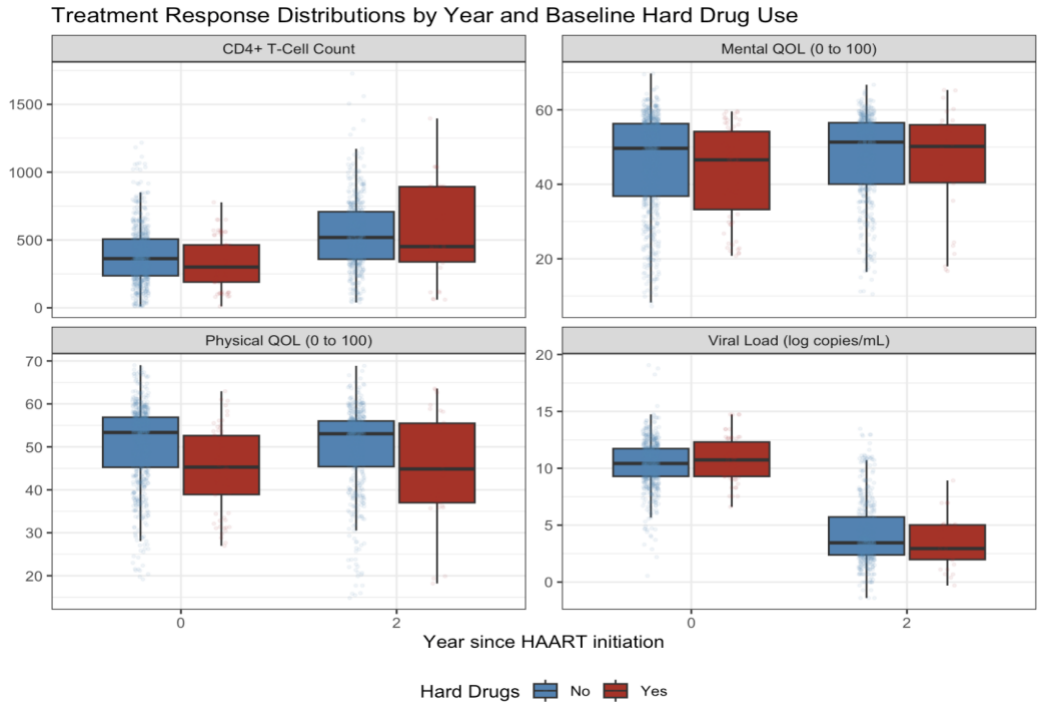

***Figure 1**. Distribution of treatment response measures at baseline (Year 0) and Year 2 following HAART initiation, stratified by baseline hard drug use. Boxplots display the median and interquartile range, with points representing individual observations.*

CD4+ T-cell counts increased from baseline to Year 2 in both groups. Participants reporting hard drug use at baseline generally exhibited slightly lower CD4 counts at both time points but with more variability at year 2. Viral load (log copies/mL) declined substantially following treatment initiation in both groups. No apparent differences in viral load are visible at both time points. Physical and mental quality-of-life scores remained relatively stable over time with drug users typically reporting lower QOL scores across both time points.

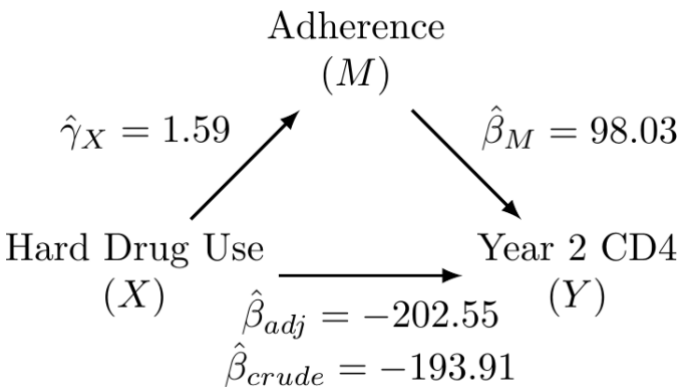

***Figure 2**. Conceptual hypothesized mediation structure linking baseline hard drug use (X), adherence (M), and Year-2 CD4 count (Y) outcome. Hard drug use may affect immune recovery directly and indirectly through its association with adherence. Covariates are not shown for simplicity.*

Baseline hard drug use (X) is hypothesized to influence Year-2 CD4 count (Y) both directly and indirectly through adherence to HAART (M). The arrow from X to M represents the association between hard drug use and adherence, while the arrow from M to Y represents the association between adherence and CD4+ T cell counts. The direct path from X to Y reflects both the crude association between hard drug use and year-2 CD4 and the adjusted association after accounting for adherence. Covariates are omitted from this preliminary diagram for clarity but are included in adjusted analyses. Initial results, show that the percent being mediated by adherence is minimal at approximately 4.4% [$\frac{\widehat{\beta_{crude}} - \widehat{\beta_{adj}}}{\widehat{\beta_{crude}}}$].